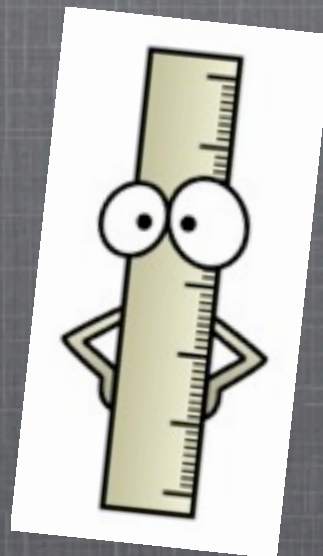


WG IX: Characteristics of Cosmological Galaxies

Some of the issues we would like to address are:

1. The relevant comparisons: code-to-code, across resolutions, with observations

2. Logistics: Cohesiveness of cosmological simulations across working groups



Registered members:

Oscar Agertz, Daniel Ceverino, María Emilia De Rossi, Fabrice Durier, **Javiera Guedes**, **Cameron Hummels**, Jason Jaacks, Dusan Keres, Andrey Kravtsov, Sam Leitner, Lucio Mayer, Daisuke Nagai, Ken Nagamine, Brian O'Shea, Joel Primack, Justin Read, Brant Robertson, Emilio Romano-Diaz, Rok Roskar, Wolfram Schmidt, Sijing Shen, Britton Smith, Robert Thompson, Matteo Tomassetti.

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Working Group XII - Interstellar Medium

Working Group XIII - Black Hole Accretion and Feedback

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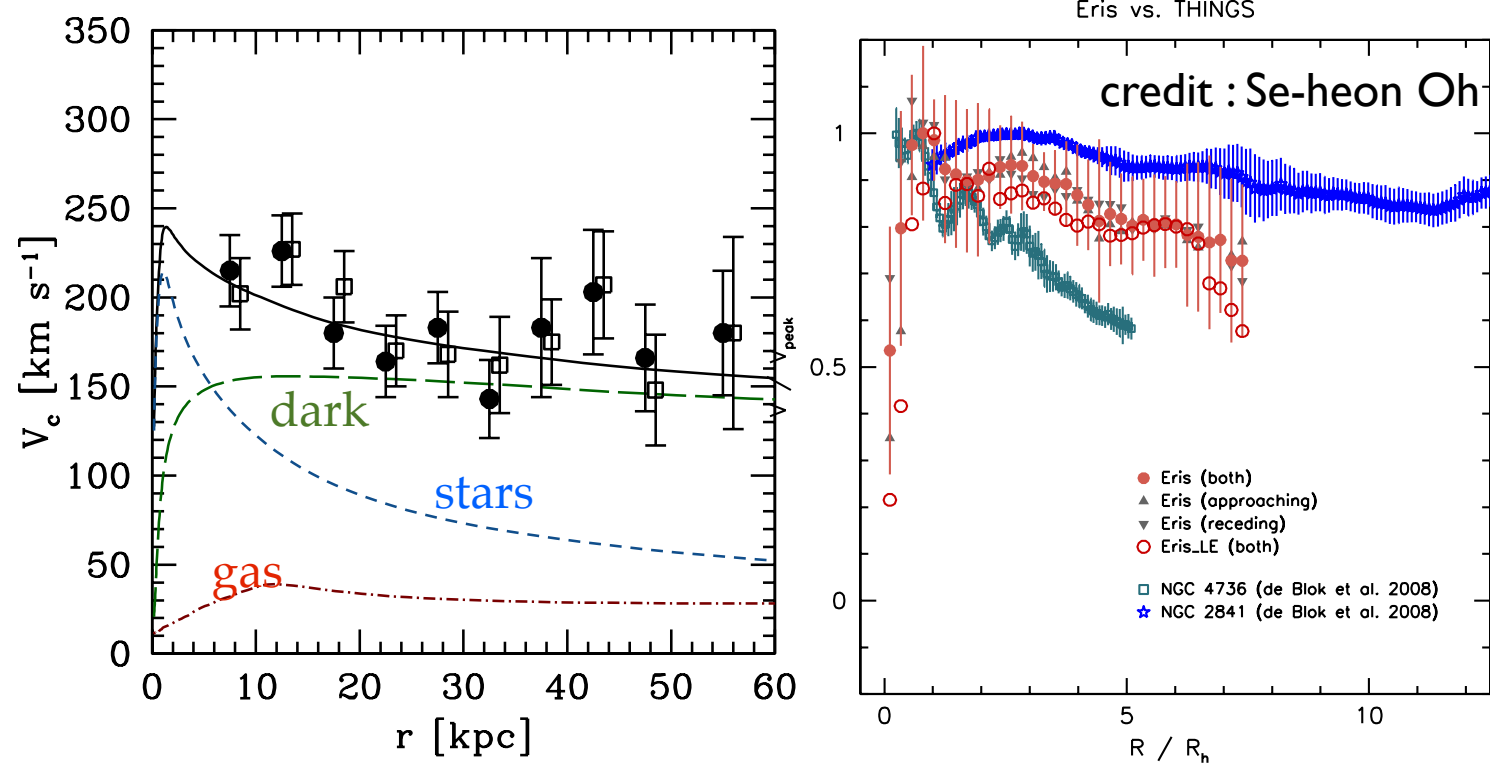
The goal of our team is to compare the properties of galaxies
with observations, across codes, and resolutions

using observer's tools whenever possible

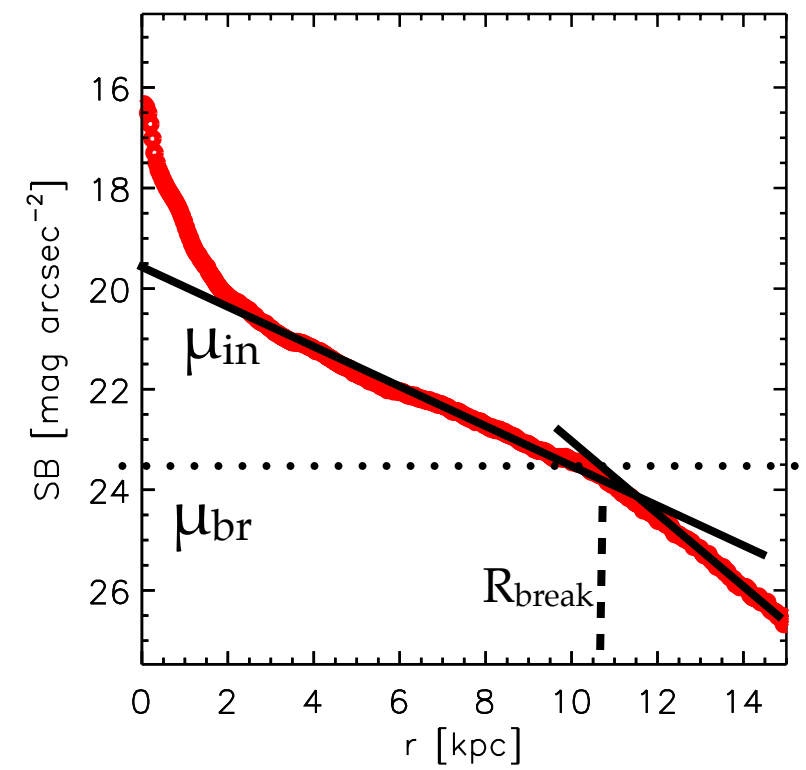
Structural Properties

Using SUNRISE, GALFIT, HI data cubes

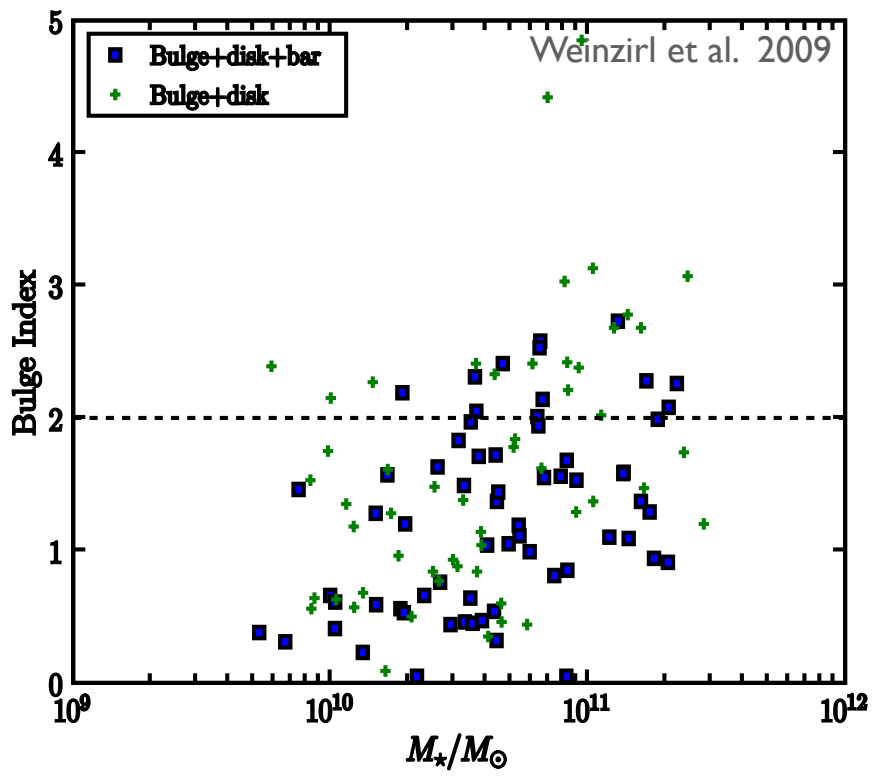
Rotation Curves



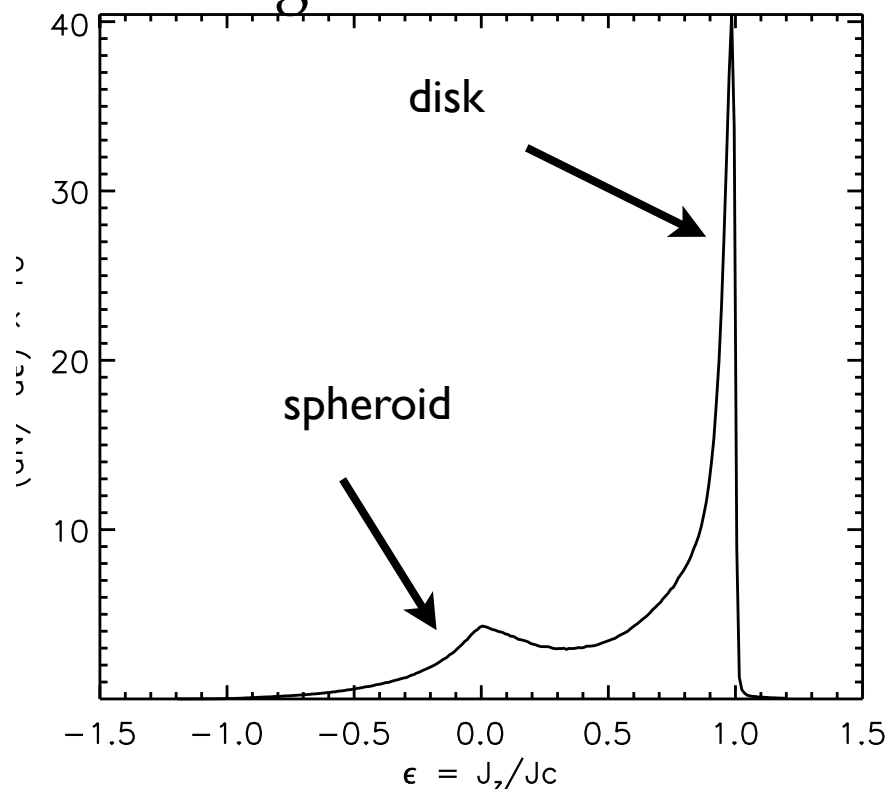
SB Profile, SB Breaks



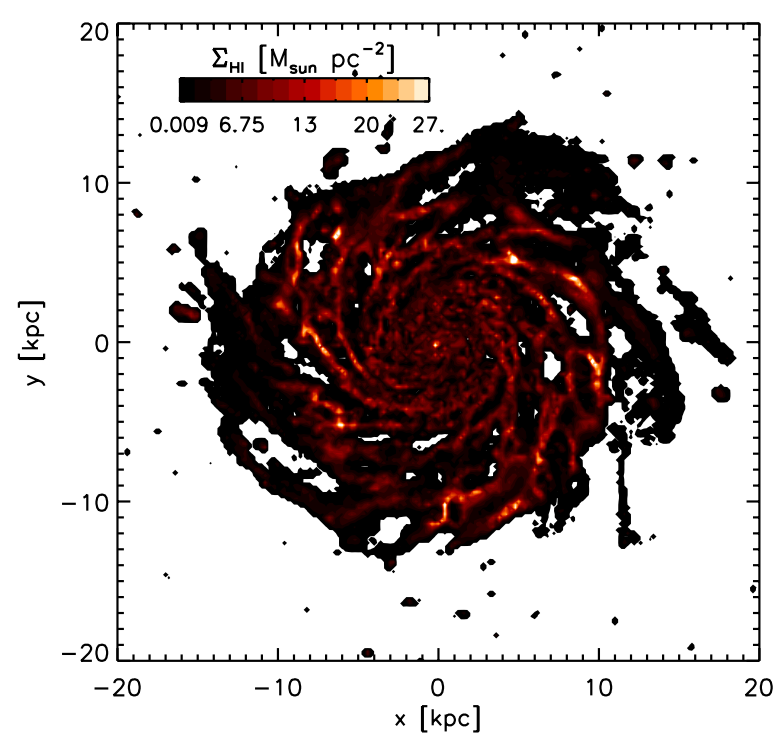
B/D ratios, n



angular momentum



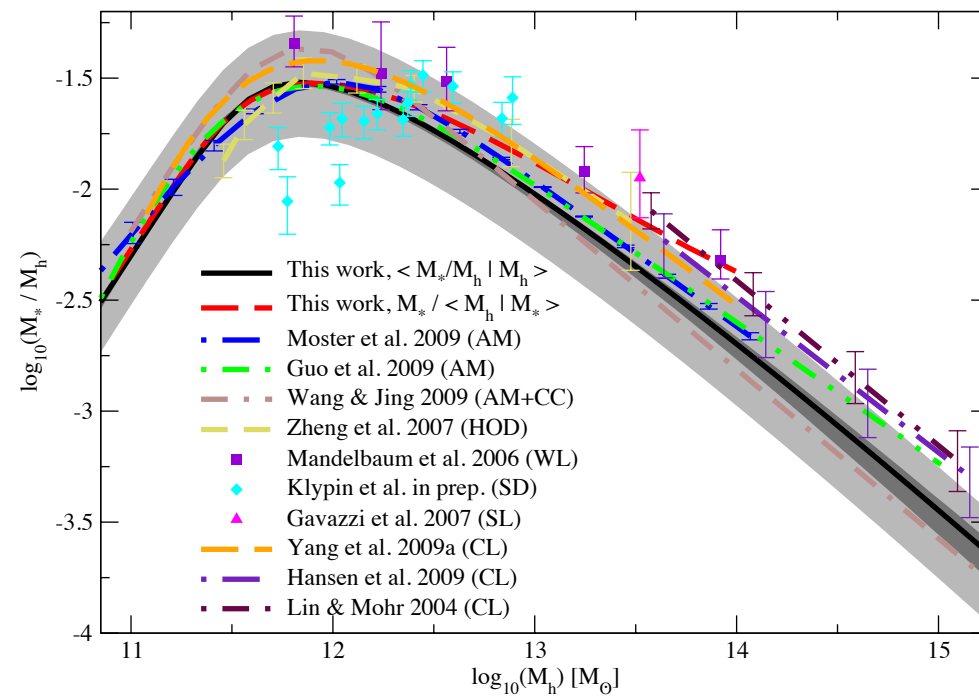
HI distribution



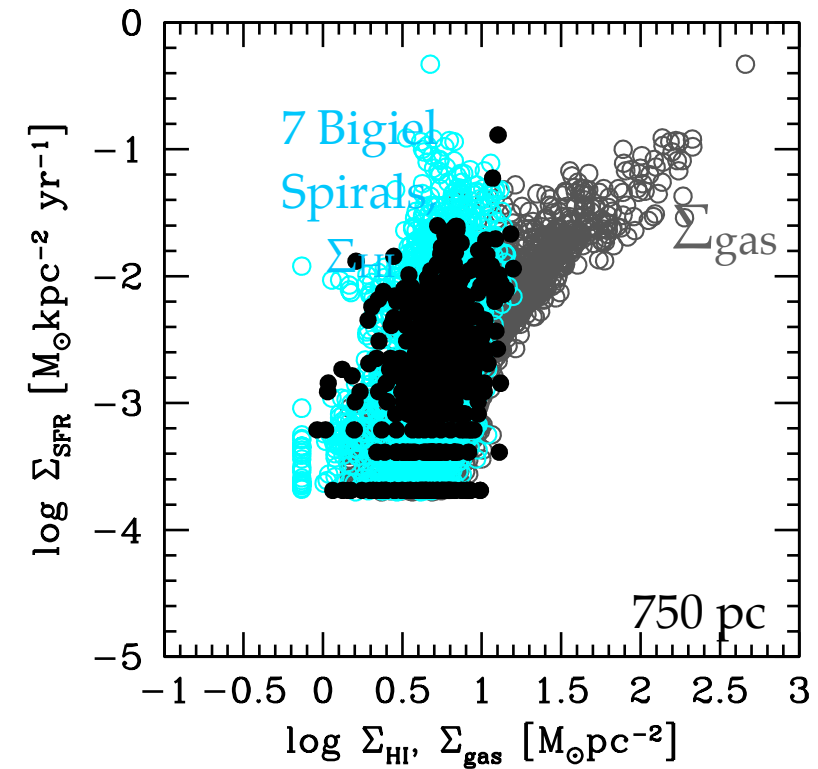
Correlation relations

Using SUNRISE, **stellar masses from SEDs?**

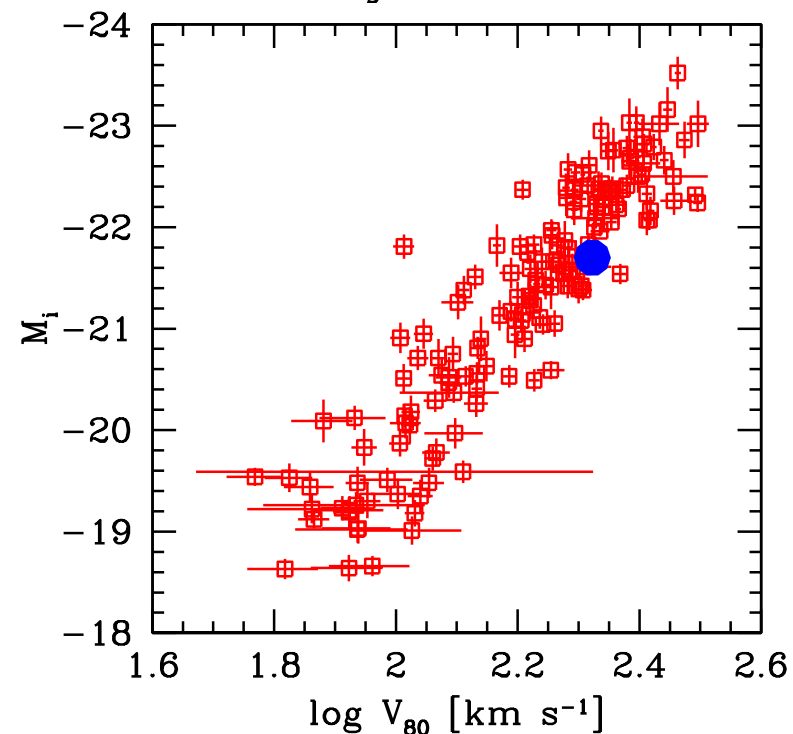
$M_{\text{halo}} - M_{\text{star}}$ Relation



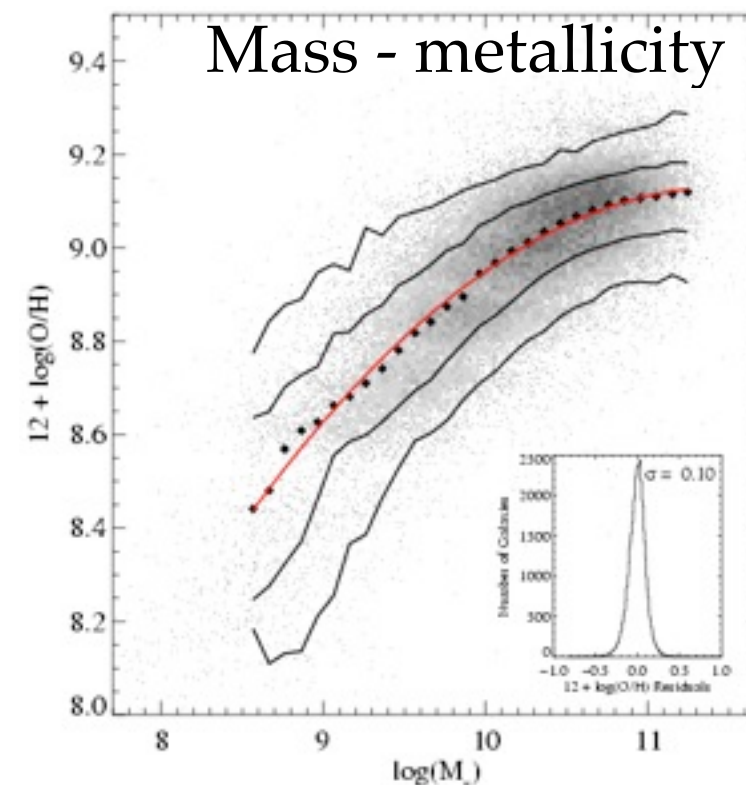
K-S Relation



Tully - Fisher



Mass - metallicity



WG IX: Characteristics of Cosmological Galaxies

Static Properties at $z=0$:

Structural properties: Rotation curves, B/D ratios (photometric & kinematic), surface brightness breaks, angular momentum distribution, gas distribution, bar strength, baryon and gas fractions, dispersion measures (electron density), X-ray emission from hot halo

Correlation relations: M^*-M_{halo} , K-S, Tully-Fisher, Mass-Metallicity, star forming sequence

Evolutionary properties:

The build-up of galactic components: SB profiles, bulge, disk, stellar halo

Star formation histories, evolution of gas fractions, disk sizes

The evolution of bars from Fourier modes or ellipsoid fitting

Stellar migration, metallicity gradients, warps (e.g. Roskar)

WG IX: Characteristics of Cosmological Galaxies

What are we trying to learn in AGORA?

- Differences between code method
- Differences between subgrid models
- Differences with observations

How can we extricate these problems?

Run simplified, low-res runs to make sure all codes agree before including high-resolution and additional physics.

WG IX: Characteristics of Cosmological Galaxies

We propose:

- What resolution to use (i.e. eventual 100 pc)
Start with 800 pc resolution and simplified physics
Check for parity (at high- z or at $z=0$ if possible)
Increase resolution by $\times 2$. Repeat.
- Common subgrid models? Leave it to each group?
Initially enforce similar subgrid models
Later expand to individual pet models.
- Comoving vs physical resolution
Follow the DM prescription in Flagship Paper?
- Repetition of runs between groups
Distribute runs between groups to avoid redundancy, designate “runners” per code, not per WG, at least for common runs

WG IX: Characteristics of Cosmological Galaxies

Possible action plan:

1. DM-only simulations (can start now), DM WG.
2. Adiabatic runs
3. Runs with simple physics (e.g. no SF)
4. Low, medium, then high-resolution cosmological runs
5. Check properties at high- z
6. Push simulations to $z=0$

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Regarding cohesiveness: do we want to analyze the same set of fiducial simulations at high and low redshift? In terms of outflows and ISM? The high- z team can manage to include very high-res sims, we can't.